FPT SCHOOL OF BUSINESS & TECHNOLOGY

AI - MACHINE LEARNING

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Hyperparameter Tuning

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Tuning Hyperparameters

- We need to tune our hyperparameters to get the best out of them.
- Hyperparameters importance are (as for Andrew Ng):
  - Learning rate.
  - Momentum beta.
  - Mini-batch size.
  - No. of hidden units.
  - No. of layers.
  - Learning rate decay.
  - Regularization lambda.
  - Activation functions.
  - Adam beta1 & beta2.
- Its hard to decide which hyperparameter is the most important in a problem. It depends a lot on your problem.
- One of the ways to tune is to sample a grid with N hyperparameter settings and then try all settings combinations on your problem.
- Try random values: don't use a grid.
- You can use Coarse to fine sampling scheme:
- When you find some hyperparameters values that give you a better performance - zoom into a smaller region around these values and sample more densely within this space.
- These methods can be automated.

Using an appropriate scale to pick hyperparameters

- Let's say you have a specific range for a hyperparameter from "a" to "b". It's better to search for the right ones using the logarithmic scale rather then in linear scale:
  - Calculate: `a_log = log(a)  # e.g. a = 0.0001 then a_log = -4`
  - Calculate: `b_log = log(b)  # e.g. b = 1  then b_log = 0`
- Ther

```
r = (a_log - b_log) * np.random.rand() + b_log
# In the example the range would be from [-4, 0] because rand range [0,1)
result = 10^r
```

It uniformly samples values in log scale from [a, b].

Using an appropriate scale to pick hyperparameters

- If we want to use the last method on exploring on the "momentum beta":
- Beta best range is from 0.9 to 0.999.
- You should search for 1 - beta in range 0.001 to 0.1 (1 - 0.9 and 1 - 0.999) and the use a = 0.001 and b = 0.1. Then:
  - a_log = -3
  - b_log = -1
- r = (a_log - b_log) * np.random.rand() + b_log
- beta = 1 - 10^r # because 1 - beta = 10^r

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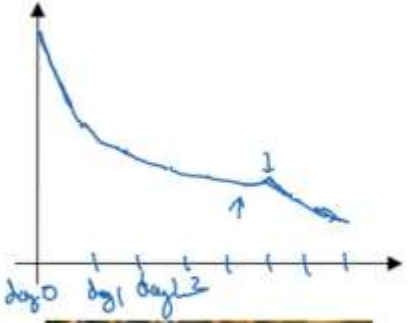
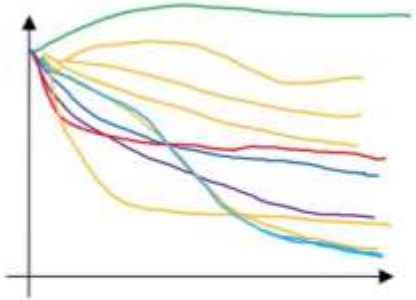
Hyperparameters tuning in practice: Pandas vs. Caviar

Babysitting one model



Panda

Training many models in parallel



Caviar

Andrew Ng

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Hyperparameter tuning in practice: Pandas vs. Caviar

- Intuitions about hyperparameter settings from one application area may or may not transfer to a different one.
- If you don't have much computational resources you can use the "babysitting model":
- Day 0 you might initialize your parameter as random and then start training.
- Then you watch your learning curve gradually decrease over the day.
- And each day you nudge your parameters a little during training.
- Called panda approach.
- If you have enough computational resources, you can run some models in parallel and at the end of the day(s) you check the results.
- Called Caviar approach.

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Practice of tuning hyperparameters

- Andrew-NG-Notes/Deep Learning Notebooks by Andrew NG/Improving Deep Neural Networks Hyperparameter tuning, Regularization and Optimization/Optimization methods.ipynb at master · ashishpatel26/Andrew-NG-Notes · GitHub

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CHALLENGE OF CLASSIFICATION 2

Option 1

https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

Option 2

https://www.kaggle.com/code/vencerlanz09/98-fire-classification-cnn-mobilenetv2/notebook

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